

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 19



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Recap of Previous Lecture



Topic

Number of onto functions

Topic

Bijjective function

Topic

Inverse of a function

Topic

Identity function

Topic

Constant function

Topics to be Covered



Topic

Identical function

Topic

Function composition





Topic : Identical function

Equal functions



Two functions 'f' and 'g' are said to be identical if and only if

- ① Domain of f = Domain of g
- and ② Range of f = Range of g
- and ③ $f(x) = g(x) \quad \forall x \in \text{Domain}$

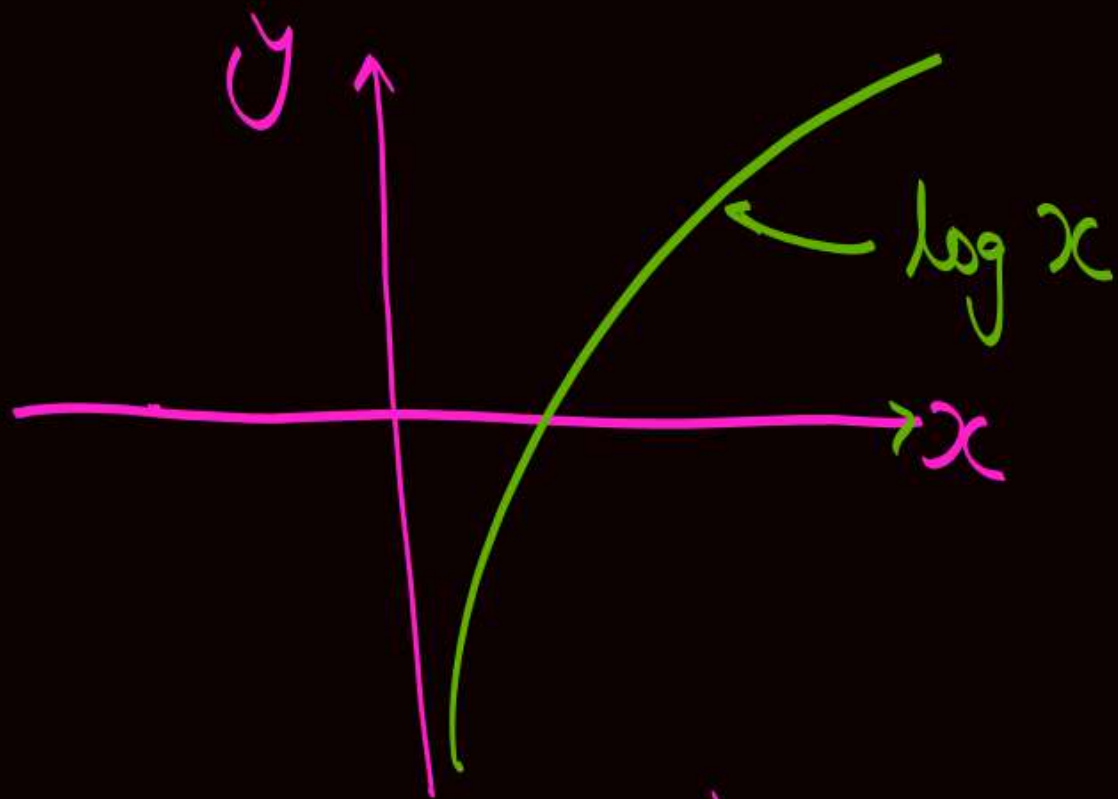
Q:- Check whether the functions 'f' and 'g' are identical or not?

$$f = \log(x^2) \Rightarrow \text{Domain of } f = \mathbb{R} - \{0\}$$

all real nos
Except '0'

$$g = 2 \log(x) \Rightarrow \text{Domain of } g = \mathbb{R}^+$$

Positive
real numbers



Domain of $f \neq$ Domain of g
 $\therefore$ 'f' and 'g' are not identical



Topic : Function composition

- Let 'f' and 'g' are two functions, Composite function of 'f' and 'g' (if exist) can be denoted by "fog"
- In general, $f \circ g \neq g \circ f$ } function Composition is not commutative
- $(f \circ g)(x) = f(g(x))$
- $(g \circ f)(x) = g(f(x))$

$$\star ((f \circ g) \circ h)$$

$$((f \circ g) \circ h)(x)$$

$$(f \circ g)(h(x))$$

$$f(g(h(x)))$$

$$(f \circ (g \circ h))$$

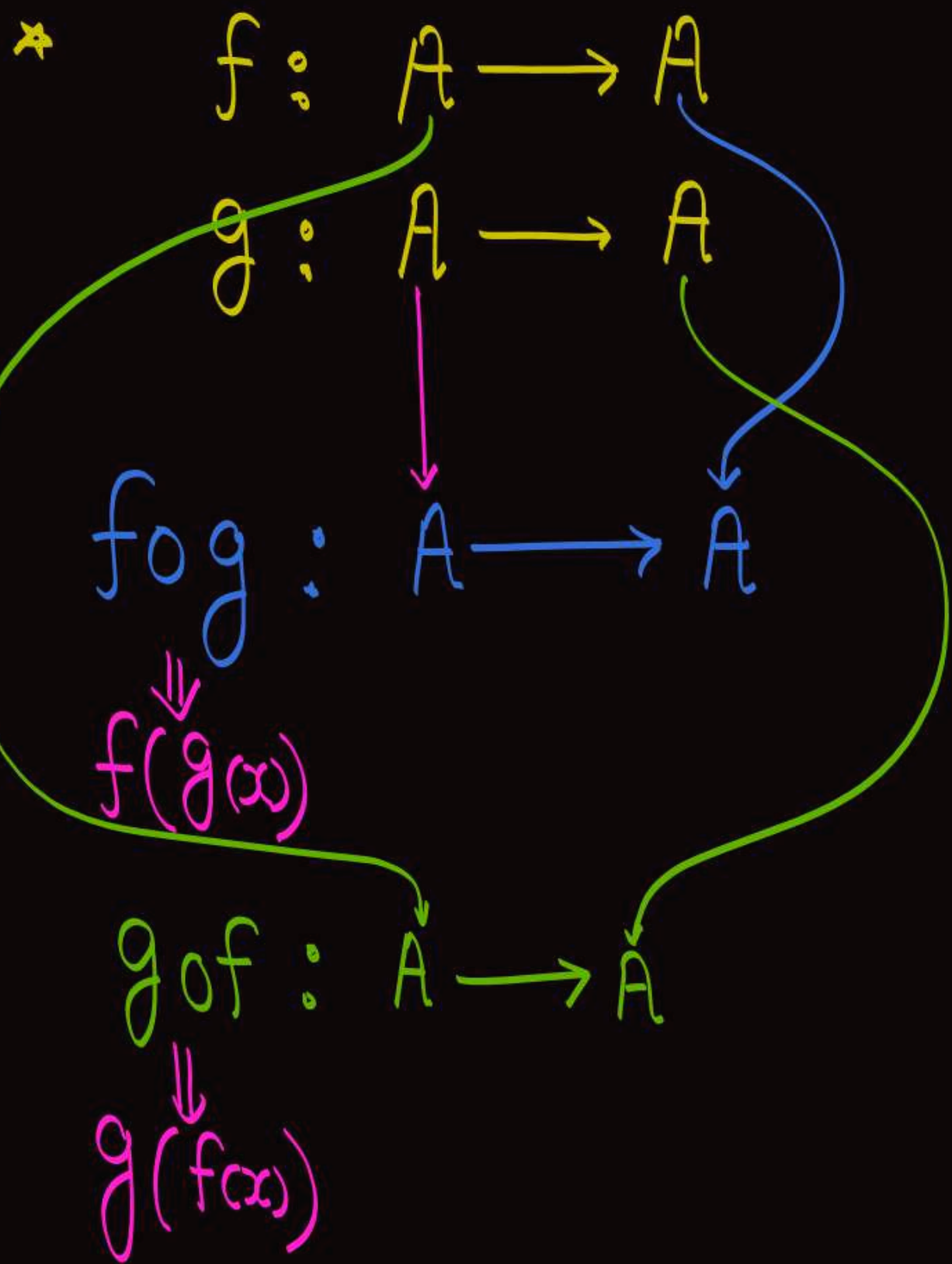
$$(f \circ (g \circ h))(x)$$

$$f((g \circ h)(x))$$

$$\equiv$$

$$f(g(h(x)))$$

Function Composition is associative } provided they
are defined }



• $f: A \rightarrow B$
 $g: B \rightarrow A$

$f \circ g: B \rightarrow B$

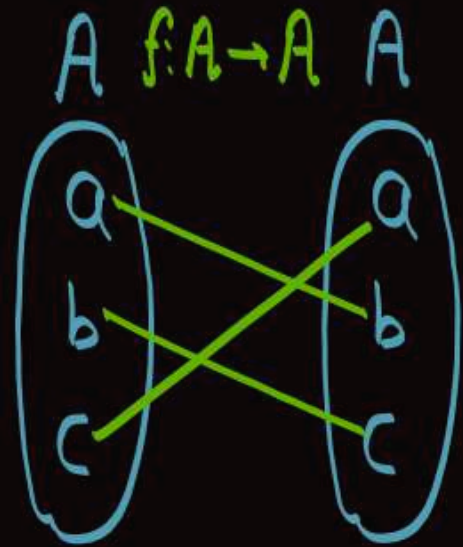
$g \circ f: A \rightarrow A$

$\rightarrow f: A \rightarrow B$
 $g: B \rightarrow C$

$g \circ f: A \rightarrow C$

$f \circ g$: Need not be defined
 $\hookrightarrow f \circ g$ will be defined if and only if
 $\text{Range of } g \subseteq \text{Domain of } f$

* Let $f: A \rightarrow A$ is a bijjective function

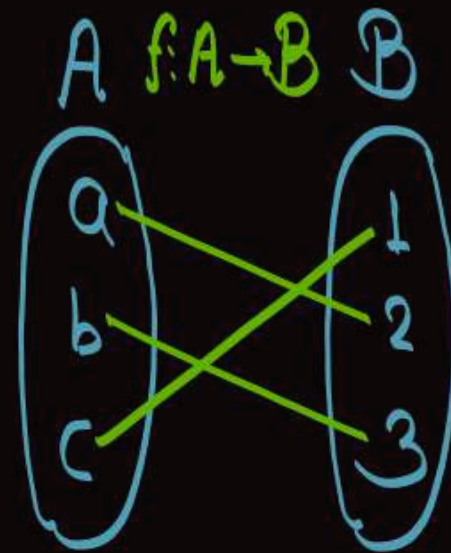


$\therefore f^{-1}$ is also defined
 $f^{-1}: A \rightarrow A$

$$(f \circ f^{-1})(x) = x \quad \left\{ \begin{array}{l} (f^{-1} \circ f)(x) = x \\ \text{Here } f^{-1} \circ f: A \rightarrow A \\ f^{-1} \circ f = I_A \end{array} \right.$$

Here $f \circ f^{-1}: A \rightarrow A$
 And $f \circ f^{-1} = I_A$

* Let $f: A \rightarrow B$ is a bijjective function



$\therefore f^{-1}$ is also defined
 $f^{-1}: B \rightarrow A$

$$f \circ f^{-1}: B \rightarrow B$$

$$(f \circ f^{-1})(x) = x$$

$\uparrow$
from set B

$$f \circ f^{-1} = I_B$$

$$f^{-1} \circ f: A \rightarrow A$$

$$(f^{-1} \circ f)(x) = x$$

$\uparrow$
from set A

$$\therefore f^{-1} \circ f = I_A$$

* Let $f: A \rightarrow A$

$$f \circ I_A = f$$

$$\begin{aligned}(f \circ I_A)(x) &= f(I_A(x)) \\ &= f(x) \\ &= f\end{aligned}$$

Similarly,

$$I_A \circ f = f$$

$$\begin{aligned}(I_A \circ f)(x) &= I_A(\underbrace{f(x)}_{\substack{\text{f(x) is a value} \\ \text{from set A}}}) \\ &= f(x) \\ &= f\end{aligned}$$

$\therefore I_A(f(x)) = f(x)$

→ Let $f: A \rightarrow B$

$$f \circ I_A = f$$

$$\begin{aligned}(f \circ I_A)(x) &= f(I_A(x)) \\ &= f(x) \\ &= f\end{aligned}$$

$$I_A \circ f = \text{Need not be defined}$$

$$(I_A \circ f)(x) = I_A(\underbrace{f(x)}_{\substack{\text{it is a} \\ \text{value} \\ \text{from set B}}})$$

$\therefore I_A$ need not be defined on value produced by $f(x)$

$$I_B \circ f = f$$

$$\begin{aligned}(I_B \circ f)(x) &= I_B(\underbrace{f(x)}_{\substack{\text{value from} \\ \text{set B}}}) \\ &= f(x) \\ &= f\end{aligned}$$

$\therefore I_B(f(x)) = f(x)$

$$f \circ I_B = \text{Need not be defined}$$

Note:- Let $*$ be any binary operation

then $(x * y)^{-1} = y^{-1} * x^{-1}$ will always hold true

does not matter whether opⁿ $*$ is commutative over the elements x and y or not

$(x * y)^{-1} = x^{-1} * y^{-1}$ may not be true in all situation

* Note: If opⁿ $*$ is commutative over elements x & y then

$(x * y)^{-1} = x^{-1} * y^{-1}$ will also hold true

Note:-



$$(f \circ g)^{-1} = g^{-1} \circ f^{-1}$$

Always true

Note $(f \circ g)^{-1}$ need not be equal to $f^{-1} \circ g^{-1}$.

because function Composition is not Commutative



Reflexive ? ✓

$$f_1(1) \leq f_1(1)$$

$$f_1(2) \leq f_1(2)$$

$$\therefore (f_1, f_1) \in R$$

$$\therefore f_1 \in F$$

$$f_1(x) = 0 \quad \forall x \in \{1, \dots, n\}$$

$$g_1(x) = 1 \quad \forall x \in \{1, \dots, n\}$$

$$\therefore (f_1, g_1) \in R$$

but $g_1(x) \neq f_1(x)$
 $\forall x \in \{1, \dots, n\}$
 $\therefore (g_1, f_1) \notin R$

Domain Codomain

Q.42 Let $\mathcal{F}$ be the set of all functions from $\{1, \dots, n\}$ to $\{0, 1\}$. Define the binary relation $\leq$ on $\mathcal{F}$ as follows:

$\forall f, g \in \mathcal{F}, f \leq g$ if and only if $\forall x \in \{1, \dots, n\}, f(x) \leq g(x)$, where $0 \leq 1$.

Which of the following statement(s) is/are TRUE?

$(f, g) \in R$ if and only if $f(x) \leq g(x) \quad \forall x \in \{1, 2, 3, \dots, n\}$

i.e; $f(1) \leq g(1)$

$f(2) \leq g(2)$

$f(3) \leq g(3)$

$\vdots$

$f(n) \leq g(n)$

Symmetric ✗

Anti-symmetric ✓

Transitive ✓

let
 Ans $f_1(x) = 1 \quad \forall x \in \{1, \dots, n\}$
 $f_2(x) = 0 \quad \forall x \in \{1, \dots, n\}$

(A) $\leq$ is a symmetric relation

(B) $(\mathcal{F}, \leq)$ is a partial order

(C) $(\mathcal{F}, \leq)$ is a lattice

(D) $\leq$ is an equivalence relation

let $(f, g) \in R$

and $f \neq g$

then

$$f(1) \leq g(1)$$

$$f(2) \leq g(2)$$

$$f(3) \leq g(3)$$

$\vdots$

$$f(n) \leq g(n)$$

if $\forall x \in \{1, \dots, n\}$
if $f(x) = g(x)$
then $f = g$

if $f \neq g$
then there exist at least one 'x' } Hence
s.t. $f(x) < g(x)$ } Anti-symmetric
in that case $g(x) \neq f(x)$
 $\therefore (g, f) \notin R$

Transitivity:

Let $(f, g) \in R$ and $(g, h) \in R$

i.e. $f(1) \leq g(1)$ and $g(1) \leq h(1)$
 $f(2) \leq g(2)$ and $g(2) \leq h(2)$
 $f(3) \leq g(3)$ and $g(3) \leq h(3)$
 $\vdots$
 $f(n) \leq g(n)$ and $g(n) \leq h(n)$

$\therefore f(1) \leq h(1)$
 $f(2) \leq h(2)$
 $f(3) \leq h(3)$
 $\vdots$
 $f(n) \leq h(n)$

$\left. \begin{array}{l} \text{i.e.} \\ f \leq h \quad \forall x \in \{1, \dots, n\} \end{array} \right\} =$
Hence
 $(f, h) \in R$
as well

$\therefore$ Transitive

Q.17

$g(.)$ is a function from A to B , $f(.)$ is a function from B to C , and their composition defined as $f(g(.))$ is a mapping from A to C .

If $f(.)$ and $f(g(.))$ are onto (surjective) functions, which ONE of the following is TRUE about the function $g(.)$?

(A)

$g(.)$ must be an onto (surjective) function.

(B)

$g(.)$ must be a one-to-one (injective) function.

(C)

$g(.)$ must be a bijective function, that is, both one-to-one and onto.

(D)

$g(.)$ is not required to be a one-to-one or onto function.



2 mins Summary



✓ **Topic**

Identical function

✓ **Topic**

Function composition

THANK - YOU